

# **When Technology Becomes Complex: Measurement, Market Structure, and Economic Value**

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## **Abstract**

This paper proposes a measure for industry-level technological complexity using a novel Network Diversity Score, which captures the connections between fundamental knowledge components. This approach outperforms other complexity metrics in key tests. The findings show that technological complexity rose steadily at both technology sector and NAICS industry levels until the Covid-19 pandemic, after which labor-intensive industries experienced a sharp decline. It is also found that the drivers of technological complexity differ between less and more competitive markets. Technological complexity is driven more by innovation incentives and financial flexibility in competitive industries, while weaker competitive pressure in concentrated industries reduces firms' incentives to invest in technological development. The results also show that patent-value measures appear to reward higher complexity more strongly, whereas stock-return outcomes respond positively to complexity only up to a certain point.

# 1 Introduction

Technology- and innovation-driven growth is widely viewed as a fundamental source of long-term economic development because it improves productivity, expands production possibilities, and improves the efficiency with which firms and economies transform input into output. With advances in technology, novel products are created, enabling more efficient production processes and stronger organizational capabilities, thereby supporting sustained gains in competitiveness and living standards. This perspective has become especially salient in recent discussions of artificial intelligence, the productivity effects of computational power may strengthen as adoption deepens and complementary investments accumulate.<sup>1</sup>

At the same time, the economic effects of innovation are unlikely to be uniform across technologies. Some innovations are relatively standardized and easy to diffuse, whereas others require integrating diverse knowledge components. This heterogeneity motivates growing interest in technological complexity, which captures the extent to which a technology reflects sophisticated, interdependent, and difficult-to-replicate combinations of knowledge. In the past literature, a universally applied method of quantifying technological complexity and its impact are still lacking, while some papers studied the links between technological complexity and economic factors, eg. [Nepelski and De Prato \(2020\)](#) links countries' positions in the global technology space to economic development, arguing that technological complexity is an important determinant of development outcomes. This paper develops a novel approach using graph theory on U.S. patent data to quantify technological complexity and compute annual complexity scores for each industry. We then compare our approach with existing measures to highlight its advantages and investigate the key drivers of technological complexity at the industry level.

Innovation continually adds new features, components, and processes to existing technologies. Over successive waves of innovation, these additional developments, building on existing ones, transform once-simple tools and mechanisms into richly elaborate processes. Ultimately, suc-

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<sup>1</sup>See Erik Brynjolfsson, "Productivity is 'Much Higher' Due to AI," *Wall Street Journal*. Brynjolfsson argues that AI's productivity effects are not limited to the direct use of AI tools; rather, they become substantial when firms make complementary investments in workflow redesign, organizational change, and human capital, allowing AI to function as a general-purpose technology.

cessive innovations drive technological complexity upward. This knowledge accumulation has long been recognized, as Newton famously wrote, “If I have seen further, it is by standing on the shoulders of Giants.”<sup>2</sup> To the best of our knowledge, the only paper that measures technological complexity from an innovation standpoint using patent data is [Broekel \(2019\)](#). His paper links technological complexity to regional growth in [Mewes and Broekel \(2022\)](#). We adopt the methodology outlines in his paper and apply to a wider application by linking it to the NAICS industry level and analyze the financial factors drives technological complexity.

Patents are the primary output of the inventive process and a well-established proxy for innovation, and prior research has long linked them to corporate valuation through measures such as patent citations, textual descriptions, and related characteristics. [Mariani, Medo and Lafond \(2019\)](#) uses the patent citation network to identify important technologies; [Kelly, Papanikolaou, Seru and Taddy \(2021\)](#) uses a text-based method to create new indicators on important patents; [Bowen III, Frésard and Hoberg \(2023\)](#) also use patent text to position startups in business cycles); and [Kogan, Papanikolaou, Seru and Stoffman \(2017\)](#) combines patent data with stock return data to calculate the financial valuation of each patent. In addition, we argue that the financial valuation may extend beyond these mechanisms, as patents are shaped by successive knowledge that increase technological complexity, which may contribute to industry-level technological complexity.

In [Arthur \(2009\)](#), technology is defined as a system of interconnected components used for practical purposes. To operationalize this concept empirically, we rely on the Cooperative Patent Classification (CPC) system, which organizes patented inventions into detailed technological categories. As discussed in [Broekel \(2019\)](#), knowledge components (10-digit CPC subclass) form the fundamental building blocks for technologies, which are usually embedded in patents as “claims” and classified under a 4-digit CPC classification system. Each patent therefore embeds a network of knowledge components. By aggregating the networks of all patents within the same 4-digit CPC class, we construct a larger technological network whose linkages among knowledge components are used to compute the Network Diversity Score (NDS), a proxy for industry-level technological

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<sup>2</sup>The metaphor “standing on the shoulders of giants” expresses that new discoveries extend prior ones. Contemporary discussions use this to emphasize that innovation almost never comes from nowhere; each advance opens new “adjacent possible” avenues for further advances. See [Farnam Street, “Standing on the Shoulders of Giants,”](#) and [Wikipedia: Standing on the shoulders of giants](#).

complexity. This measure is based on four structural network indicators, each capturing a distinct property of the technological network: (1) module density - the number of densely connected groups relative to the overall size; (2) variation in module size - how different in the size of the densely connected groups are from each other; (3) overall network connectivity - how well the nodes are connected with each other; and (4) the growth of small subgroups within the network - which compares the number of patterns with three nodes to the number of patterns with four nodes.

This approach of measuring technological complexity outperforms other existing complexity measures in two important ways. First, when comparing with other existing complexity measures that do not use network theory, this method performs better in terms of its linkage to R&D intensity (Broekel (2019)). Since, long-term corporate valuation is critically dependent on R&D investments and R&D intensity (see, Coad and Rao (2008)), we expect technological complexity to be an important determinant of corporate valuation. Second, the proposed measure, instead of relying on a single network feature, combines four different structural indicators at once, providing the ability to classify networks by appropriately capturing the degree of heterogeneity of the interconnections between nodes. This is proved by comparing it to three standard network measures – the Bertz index, Wiener index, and mean distance deviation (MDD). In environmental technologies, Bertz, Wiener, and NDS all trend upward steadily, whereas MDD is noisy with little informative pattern. In AI, NDS clearly reflects the post-2021 surge in complexity consistent with the recent boom, while among the other metrics, only the Wiener index shows a similar but weaker uptick. Overall, the evidence indicates a sustained performance for NDS relative to the other methods.

The primary contribution of this paper lies in establishing the linkages between technological complexity, product market share and financial market valuation. Because technological knowledge components, as captured by CPC codes, form the foundation of our complexity measure and are likely to be shared within sectoral and industry categories, we conduct our analysis at the NAICS industry level. We then examine how industry-level technological complexity relates to firms' operating characteristics and patent activity. Our main hypothesis is that firms' incentives to pursue technological complexity depend on the competitive environment in which they operate.

The determinants differ substantially across industries with different levels of product-market con-

centration, measured by HHI. In more competitive industries, firms face greater pressure to differentiate themselves and protect their market position. When profitability is relatively low, this pressure may strengthen incentives to invest in new and more complex technologies, while higher market valuations can provide both stronger growth expectations and greater capacity to finance risky, long-horizon innovation. By contrast, in more concentrated industries, competitive pressure is weaker and the benefits of continuous technological differentiation may be smaller. In these settings, larger firms may be better positioned to sustain complex technological portfolios because of their greater organizational capacity, stronger incentives when lower-valued, and accumulated technological capabilities.

Technological complexity is strongly related to firm valuation, and establishing this relationship is one of the main contributions of this paper. The results suggest that complexity is not only an indicator of the technological structure of an industry, but also an economically meaningful signal for investors and market participants. Firms operating in more complex technological environments tend to be associated with higher patent value and stronger market responses, especially when complexity remains within a range that investors can evaluate and understand. At very high levels, however, complexity may also create uncertainty about commercialization, implementation costs, and future profitability, which weakens the positive market response. Overall, the evidence shows that technological complexity is closely connected to how firms are valued, suggesting that the structure of technological knowledge plays an important role in shaping firm value.

One of the major contributions of this paper is developing a two-step conversion process and associated concordance file to merge our CPC-based complexity scores with NAICS-classified economic and financial data. First, CPC codes are mapped to the initial three NAICS versions using the ALP probabilistic links from [Nathan, Lybbert and Jason \(2019\)](#). Since our study spans seven NAICS revisions, we then chain together adjacent NAICS versions using probability weights from the Census Bureau's concordance files. Because complexity scores must stay on the same scale during conversions and the patent/citation count should be preserved at an aggregated level, separate weighting schemes and formulas are applied for each measure – details are provided in the Data Appendix. This conversion framework is a key contribution of this analysis, as it can be applied to translate any CPC-based variable to NAICS industries across multiple NAICS versions.

After mapping patent data and their corresponding CPC codes to NAICS industries, we first look at the distributions of complexity scores, which show that overall complexity rises, with more technologies scoring high and fewer scoring zero, and this fits the cumulative nature of innovation. Breaking the trend down by sector shows a steady increase across all areas until 2020, followed by a post-2020 pullback that is most pronounced in fixed construction. Besides, we also find that sectors with fewer usable observations have limited innovation opportunities and thus generate less patents. By focusing on 8 selected industries, we find that the post-pandemic decline is concentrated in labor-intensive fields, while high-tech industries see no significant fall. Also, the industry-level graph also shows that complexity scores are not comparable across industries, making comparisons and analysis at the industry level challenging. To address this, we standard the score by de-meaning it within each industry, generating an “excess” complexity score that is comparable across all industries.

The rest of this paper is organized as follows. Section 2 explains how we construct the technological complexity score, compares it with other measures, and describes the conversions to and between NAICS versions. Section 3 presents the trends and key properties of the complexity scores over time. Section 4 details our data and regression models and explores the drivers of complexity across five sample splits. Section 5 shows how does technological complexity relate to firm valuation. Finally, Section 6 offers our conclusions.

## **2 Construction of Complexity Score**

### **2.1 Idea**

In network theory, networks are broadly classified into three types: ordered networks, which exhibit predictable connections between nodes and present low heterogeneity; random networks, in which the connections are determined purely by chances, so the heterogeneity degree is the largest; and complex networks, which lie between those extremes, is a mixture of ordered & random networks. Building on this classification, the industrial complexity score in this paper is calculated

using the Network Diversity Score (NDS) — initially proposed by [Emmert-Streib and Dehmer \(2012\)](#) in a biological context. Unlike other network-based metrics, NDS simultaneously uses four structural indicators, enabling it to distinguish between ordered, complex, and random networks.

[Emmert-Streib and Dehmer \(2012\)](#) originally proposed NDS as a purely theoretical metric to quantify the complexity of a network. [Broekel \(2019\)](#) later adapted this measure to the patent world. Technologies have been viewed as combinations of interconnected elements ([Arthur \(2009\)](#), [McNerney, Farmer, Redner and Trancik \(2011\)](#)). These elements — ranging from physical parts and functional modules to individual steps in an industrial process — are considered linked whenever a change in one alters the behavior or performance of another. In [Broekel \(2019\)](#), a technology is described as a recipe, in which the information includes the knowledge components and the combination of how these components are combined. For example, the same basic inputs (flour, eggs, milk, oil) can produce something as simple as scrambled eggs or something as elaborate as a multi-layer cake. Metaphoring knowledge components as ingredients and technologies as recipes illustrate that complexity depends less on the ingredients themselves and more on how knowledge components are linked with each other, specifically on the heterogeneity of their connections. By applying NDS to technological networks, we can rigorously quantify how diversely knowledge components interact.

## 2.2 Construction

By modeling each 4-digit CPC code as a network, the NDS represents the diversity score of the networks, and each technology (4-digit CPC code) is regarded as a combination of knowledge components ([Broekel \(2019\)](#)). Therefore, the complexity of each depends on how these knowledge components are linked to each other. The more diverse and complex the direct or indirect connection, the more complex this technology is. The author considered the lowest level of CPC classes (10-digit CPC subclasses) as knowledge components and their co-occurrence on patent as linkages ([Fleming and Sorenson \(2001\)](#), [Sorenson and Fleming \(2004\)](#)).

The data files used to construct the networks are from USPTO’s PatentsView as of July 2024. For

each 4-digit CPC code, a network is built annually using all patents classified into this 4-digit CPC class with a moving window of 3 years. The nodes are the 10-digit CPC subclasses belonging to this 4-digit class and have existed in the patent files of the time range. The edges between two subclasses are defined as existing if these two subclasses ever co-occurred in one of the patent files. For each 4-digit CPC class, we first build its network and extract the giant component — the largest connected cluster of nodes and edges. Following [Broekel \(2019\)](#), we then randomly pick 50 nodes from that component. From each chosen node, we run a 150-step random walktrap to generate a subnetwork. Computing the iNDS on each of these 50 subnetworks by using four structural indicators and averaging the results gives the annual complexity score for the CPC class. The equations for iNDS and NDS are the followings with  $S = 50$ :

$$iNDS(G_T) = \frac{\alpha_{module} \cdot r_{graphlet}}{v_{module} \cdot v_{\lambda}} \quad (1)$$

$$NDS(\{G_T^S|G_M\}) = \frac{1}{S} \sum_{G_T \in G_M}^S iNDS(G_T) \quad (2)$$

$$Score(G_M) = -\log NDS(\{G_T^S|G_M\}) \quad (3)$$

The four network structural indicators in equation 1 represent different properties of a network.  $\alpha_{module} = \frac{M}{n}$  with  $M$  being the number of modules (densely connected subgroups) and  $n$  being the number of nodes. This measurement represents the module density of a network, namely, how many densely connected groups there are compared to the overall size. It gives a higher value for a less heterogeneous network and helps distinguish between ordered, complex, and random networks.  $v_{module} = \frac{var(m)}{mean(m)}$  with  $m$  being a vector containing sizes of all modules, showing how different in size the modules are from each other. This variable mainly distinguishes the networks within the three types of networks and gives higher values for higher heterogeneity.  $V_{\lambda} = \frac{var(\Lambda(L))}{mean(\Lambda(L))}$  with  $\Lambda(L)$  representing the eigenvalues of Laplacian matrix for the network tells how well a graph is connected. So  $V_{\lambda}$  is also higher for more heterogeneous networks within the three types of networks. The last indicator  $r_{graphlet} = \frac{N_{graphlet(3)}}{N_{graphlet(4)}}$  and  $N_{graphlet(a)}$  is number of graphlets of size  $a$ . This captures how do small subgroup in the network growth, or compares the number to patterns with three nodes to the number of patterns with 4 nodes. It helps to classify

networks into the three types since  $r_{graphlet}$  is smallest for random networks, largest for ordered networks and moderate for complex networks.

Figure 1 shows the changes in complexity score at the level of 1-digit CPC industry by averaging the 4-digit level score. The complexity score shows a tendency of increase across years for all industries before 2020. In contrast, decreases occur in all industries after the pandemic, and the sharpest drop happens in Fixed Constructions. This is mirrored by an increase in work-from-home cases and a decrease in the need for transportation and office buildings. Considering the fact that patent application is a revenue-driven activity from the perspective of firms, uncertainty about future policies and market demand might cause fewer patents to be applied and the technological complexity score to decline correspondingly.

The change in the big-picture of complexity score is clearer in figure 2. It shows the distribution of the industrial complexity scores of almost 672 technologies every year from 1982 to 2024. The shading darkens as time goes on. For each year, the distribution is bimodal with a peak at zero and bell-shaped at moderate values. The peaks at zero are caused by the fact that many technologies have too few patents to construct a complicated network and to calculate a score. As years pass, growing patent numbers lead the density at zero to become lower. Furthermore, the maximum density moving to the right side is consistent with technologies being built upon existing ones. This figure shows a similar trend with EU technologies' changes as described in [Broekel \(2019\)](#) using EU patent data.

## 2.3 Comparison

The advantages of NDS become clear both when compared with traditional complexity measures that ignore network topology and in head-to-head comparisons with other network-based metrics, where it consistently outperforms.

The first part was already discussed and studied by Dr. Broekel in [Broekel \(2019\)](#). The author carried out a direct comparison of NDS against Fleming's modular complexity (FS modular; [Fleming and Sorenson \(2001\)](#)) and Balland's technological complexity index (KCI; [Balland and Rigby](#)

(2017)) to see how each relates to R&D intensity. NDS correlates positively with patent counts and with collaborative R&D (measured by inventor numbers), assigns higher scores to high-tech fields in every period, and gives older technologies lower complexity scores. In contrast, FS modular increases complexity for older technologies, which contrasts the stylized fact. KCI not only relates negatively to patenting and collaborative R&D but also ranks older technologies as more complex. This comparison highlights that of these three, the structural diversity method (NDS) reflects certain features commonly associated with technological complexity.

The comparison between structural diversity (NDS) and other network-based metrics is theoretically mentioned in Emmert-Streib and Dehmer (2012). Other 16 network complexity measures are also studied in Emmert-Streib and Dehmer (2012) and compared with NDS. These measures differ from NDS in two main ways. First, NDS combines four distinct structural indicators, whereas the others each rely on a single principle. Second, it constructs the estimation by sampling a finite set of networks from the population and applying the central limit theorem, which helps to eliminate ambiguity from one simple network. The authors contended that NDS is the only system capable of distinguishing ordered, complex, and random networks unambiguously by capturing the heterogeneity in how nodes are connected. In this paper, three traditional network diversity measures are applied to patent data and evaluated alongside the NDS. The three measures are the Bertz index, Wiener index, and mean distance deviation.

Bertz index expresses the total structural information content of a graph and takes the formula (Bertz (1983)):

$$B(G) = 2n \log(n) - \sum_{i=1}^k |N_i| \log(|N_i|), n = \text{number of nodes} \quad (4)$$

Wiener index is one of the most widely used topological indices in graph theory. It captures the overall compactness by summing up the pairwise length of the shortest path between nodes. The formula for this measurement is (Wiener (1947)):

$$W(G) = \sum_{\{u,v\} \subset V} d(u,v), d(u,v) = \text{distance between nodes } u \text{ and } v \quad (5)$$

Mean distance deviation quantifies how spread out the pairwise shortest-path distances are around their average. It complements the Wiener index (which sums distances) by capturing dispersion rather than the total sum using the distances between nodes. In consequence, it offers insight into heterogeneity, information that neither the Bertz index nor the Wiener index can provide. It is defined as (Skorobogatov and Dobrynin (1988), Todeschini and Consonni (2008)):

$$MDD(G) = \frac{1}{n} \sum_{i=1}^n \left| \mu(v_i) - \frac{2W(G)}{n} \right| \quad (6)$$

where

$$\mu(v_i) = \sum_{j=1}^n d(v_i, v_j) \quad (7)$$

When complexity over time for two areas – environmental area (Figure 3) and AI tech (Figure 4) – are reported, apparent differences can be shown. In the environmental sector, the structural diversity score, Bertz index, and Wiener index all follow a similar, steadily increasing path, while the mean distance deviation (MDD) jumps around with no clear pattern. In contrast, for AI technologies, the structural diversity score rises gently up to 2020 and then spikes up sharply from 2021 onward, matching the recent AI boom. Of the other three metrics, only the Wiener index shows a noticeable uptick after 2021 and a gradual rise before that.

The contrasts become even clearer when we aggregate the data up to the 1-digit CPC level (Figure 1 for structural diversity score). The Wiener index (Figure 5) and MDD (Figure 6) stay flat over the years, and their scales differ broadly across sectors to be comparable. The Bertz index (Figure 7) does increase across every sector, but is not yet comparable, because the scores in the electrical sector significantly exceed the rest.

## 2.4 Conversion to NAICS

The classification system used by the USPTO’s dataset is the Cooperative Patent Classification (CPC). When calculating the industrial level complexity score using the patent file from USPTO, the classification is documented under CPC, so one score is generated for each four-digit CPC

code in each year. Conversion to other forms of classification is needed since the CPC is used to manage patent documents. The North American Industry Classification System (NAICS) is the standard code used by federal agencies to classify business establishments for the purpose of analyzing statistical data related to the U.S. business economy. Therefore, in order to link the industrial complexity score with Compustat and CRSP data, the score needs to be converted from the CPC to the NAICS form. The detailed conversion processes for the complexity score can be found in Appendices [A.1](#) and [B.1](#).

The North American Industry Classification System (NAICS) was first adopted in 1997 to replace the Standard Industrial Classification (SIC) system. Since then, NAICS has been scheduled to be reviewed every five years for potential revisions, and will be for the foreseeable future, so that it can keep pace with the emerging industries and make clarifications for select industry definitions. Since the time range in my dataset is 1985 - 2023, the relevant NAICS versions include NAICS 1997, NAICS 2002, NAICS 2007, NAICS 2012, NAICS 2017, and NAICS 2022. Historical NAICS codes (naicsh) from Compustat, which represent each firm's NAICS classification for each fiscal year, were used, applying the relevant NAICS version for that year. The crosswalks from CPC to NAICS are only feasible for the first three NAICS versions, which are the NAICS 1997, NAICS 2002, and NAICS 2007, so the conversion between different versions of NAICS is also necessary. In this paper, two types of conversion will be applied. If the analysis needs to link the industrial complexity score with Compustat, conversion to the corresponding version will be applied. If only comparison within complexity score data across years is needed, conversion to the NAICS 2007 version is exercised for easier interpretation.

### **3 Properties**

From this section on, the complexity score will be shown at 4-digit NAICS level by averaging the affiliated 6-digit NAICS codes' scores. This conversion is necessary because further analysis will be based on industry-level by aggregating firm-level data from Compustat or CRSP, and the firm number categorized in each 6-digit NAICS industry is limited in each fiscal year. In order to

increase the research power at the industry level, 4-digit NAICS are the better choice because they include more firms each year, facilitating a more comprehensive industry-level view.

Figure 8 shows the distribution of the complexity scores at the level of 4-digit NAICS, in which the outlier represents the year 2024 with incomplete granted patent data. It shows a similar trend with figure 2; majority density moves to the right side, and the left tail becomes thinner, indicating that the overall technologies are becoming more complex over the years. The disappearance of peaks at zero is caused by the conversion from CPC to NAICS, and there are few cases where there are not enough patents to calculate for all 4-digit CPC classes one NAICS code mapped to.

Compared to figure 8, which presents the change in the technological complexity score on a larger scale, figure 9 selects eight industries and shows the annual complexity scores and employment for each industry. They are, from left to right and top to bottom: Agriculture Group; Air and Rail Transportation; Computer and Electronic Product Manufacturing; Data Processing; Motor Vehicle Manufacturing; Oil and Gas Extraction; Software; and Textile Mills. The chosen industries, ranging from labor-intensive to high-tech areas, all exhibit climbing scores until 2020. After the Covid-19 pandemic struck, labor-intensive industries suffered reductions in complexity scores, while high-tech ones remained stable. Given that labor is one of the main input capital resources, in order to find out whether the drops in labor-intensive industries are paralleled with a decrease in employment, employment data from the Bureau of Labor Statistics's OEWS dataset is used and represented by red lines in the graphs. It turns out that the employment statistics remained steady for these industries and can not account for the drop in complexity scores.

Another observation from Figure 9 is that the complexity score varies across industries, making direct comparisons challenging. For example, the Agriculture Group industry has scores ranging from 12.79 (in 1984) to 33.82 (in 2018), while the Data Processing industry spans from 25.60 (in 1984) to 46.97 (in 2020). High-tech industries' technological complexity levels from forty years ago are estimated to be similar to those of labor-intensive industries today (Agriculture Group scored 25.90 in 2023). This difference arises because each industry is inherently distinct, and the initial scoring system captures this because of the variances in the patent data. Consequently, it is necessary to establish an industry-specific benchmark to enable comparisons of technological

complexity scores across industries. Each benchmark is determined by averaging the industry’s scores over the entire time period, and the excess score is calculated by subtracting this benchmark from the initial complexity score. Figure 10 presents the same eight industries’ excess scores from 1982 to 2023, and after adjusting by subtracting the benchmark, the scores become more comparable across industries.

Table 1 summarizes excess score statistics at the 2-digit NAICS level (the system’s broadest category). Column 1 is the 2-digit NAICS sector. Column 2 reports, for each 2-digit NAICS sector, the total number of panel observations in the data set from 1982 to 2023. Column 3 shows the total count of 4-digit NAICS codes according to the Census Bureau, and Column 4 reports how many of those have valid excess scores over the same period. Finally, Columns 5 and 6 present the median and standard deviation of the excess scores for each sector. There are substantial gaps between Columns 3 and 4 in several sectors, most notably 42 (Wholesale Trade), 48 (Transportation), 53 (Real Estate and Rental and Leasing), and 81 (Other Services). In these industries, fewer 4-digit codes have measurable complexity scores because they generate relatively few patents – reflecting inherently limited opportunities for innovation, either because technological improvement isn’t central to their processes or because they’ve largely reached an innovation plateau. Figure 11 presents a scatter plot of each 2-digit NAICS sector over time, serving as a graphical counterpart to Table 1. The consistent upward trends in every sector further confirm that technological complexity has increased over the years.

## 4 Drives of Complexity

### 4.1 Data

In this section, regressions are used to discuss the determinants of complexity at the industry level (4-digit NAICS). Patent activity are incorporated by counting, for each CPC code in year  $t$  quarter  $q$ , the number of patents filed (“*patent\_count*”) and the citations they received within one, two, and three years of filing (named “*citation\_1year*”, “*citation\_2years*”, and “*citation\_3years*”, respec-

tively). These data come from the USPTO’s PatentsView database. The annual growth rates of those measures are calculated, along with the three-year growth rate of patent count. In the regressions, lagged versions of these variables are employed. For example, “*L.citation\_1year*” in year  $t$  quarter  $q$  measures the one-year citations accrued by patents filed in year  $t - 1$  quarter  $q$  (up to the end of year  $t$  quarter  $q$  for patents filed at the end of  $t - 1$ ), and similarly for “*L2.citation\_2years*” and “*L3.citation\_3years*”, so that all patent measures reflect prior activity when explaining current complexity.

For this section, all analyses will be shown and discussed in conversion to the corresponding NAICS version since the patent-related data (complexity score, patent count, and citation count) need to be linked with Compustat. For the data within 1985-2011, the ALP probabilistic linkages from [Nathan et al. \(2019\)](#) are used to map data to the relevant NAICS version. For 2012–2023 – when later NAICS versions lack direct CPC links – data are first converted NAICS 2007 first using the CPC-NAICS07 linkages, then, conversions between adjacent NAICS versions are applied. The full details on converting patent and citation counts from CPC to NAICS and between adjacent NAICS versions are in Appendices [A.2](#) and [B.2](#), respectively. Once patent data are mapped to 6-digit NAICS codes, we roll them up to the 4-digit level by summing the values for all 6-digit codes that fall under each 4-digit industry.

Compustat provides quarterly firm-level accounting data along with each firm’s historical 6-digit NAICS code. The accounting variables considered include total assets, net income, leverage ratio, share of sales, R&D expenditure, external finance dependence, and the Herfindahl-Hirschman Index (HHI). The quarterly accounting figures are then obtained at the industry level by taking the median of all affiliated firms. A panel of accounting data for each 4-digit NAICS ranging from 1985 (the first year with historical 6-digit NAICS records) to 2023 was constructed. Then a one-to-one merge with excess complexity scores and patent activity indicators was performed to prepare for the regressions.

Table [2](#) and Table [3](#) provide definitions and summary statistics for all potential explanatory variables. The statistics show each variable in its original units and in its natural-log form for use in the regressions. To address potential multicollinearity among variables, a correlation matrix is also

presented in Table 5 for the key variables used in our subsequent regressions. As expected, there are weak relationships between accounting variables. Likewise, accounting measures and patent indicators generally show weak or no linear relationships, except for the HHI, which will be used to split the dataset. Among the patent indicators, the count measure shows a weak correlation with the growth measure.

## 4.2 Models

Based on the correlation matrix, the base regression model is set to be as Equation 8 to avoid the multicollinearity risk, in which, the dependent variable  $C_{i,q}$  is the natural logarithm of the excess complexity score at quarter  $q$  for the industry (4-digit NAICS)  $i$ . Three logged accounting variables are included: total assets, net income, leverage ratio, and R&D expenditure, all with lagged terms.  $g(patent)_{i,q-1}^1$  is the logged growth rate of the patent count at quarter  $q - 1$  (the growth rate of patent from quarter  $t - 2$  to quarter  $t - 1$ ), and  $Citation_{i,q-5}^1$  is the logged number of citations the patent filed in quarter  $q - 5$  received within one year after filing (up to the end of quarter  $q - 1$ ).

$$C_{i,q-1} = \beta_{i,q}^0 + \beta_{i,q}^A \cdot Asset_{i,q-1} + \beta_{i,q}^{ROA} \cdot ROA_{i,q-1} + \beta_{i,q}^L \cdot Leverage_{i,q-1} + \beta_{i,q}^{MB} \cdot MB_{i,q-1} \\ + \beta_{i,q}^g \cdot g(patent)_{i,q-1}^1 + \beta_{i,q}^C \cdot Citation_{i,q-5}^1 + \gamma_i + \alpha_t + \sigma_q + u_{i,q} \quad (8)$$

## 4.3 Results

Table 6 summarizes five regressions based on different sample splits. In the full-sample results, accounting characteristics are not significantly associated with industrial technological complexity. By contrast, both indicators of patent-activity are significantly related to technological complexity. A 1% increase in patent growth is associated with approximately a 0.038% increase in the excess complexity score. This positive relationship is consistent with the idea that industries experiencing faster technological development also tend to exhibit more complex technological networks. The number of citations received represents the scientific and economic influence of a patent (Kogan et al. (2017), and Arora, Belenzon, Ferracuti and Nagar (2024)). Aggregating citations to

the industry level therefore captures the extent to which inventions within an industry influence subsequent technological development. The results show that a 1% increase in one-year citation counts is associated with approximately a 0.064% decline in the complexity score. One possible explanation is that highly cited patents may be built around widely adopted or standardized technological components. Subsequent inventions build on these patents, generating high citation counts while simultaneously producing more homogeneous technological networks. In this sense, citation counts capture the influence and diffusion of existing technologies, whereas the complexity measure is more closely related to the diversity and recombination of technological components.

After splitting the sample at the median Herfindahl–Hirschman Index, Table 6, columns (2) and (3), show that the determinants of technological complexity differ substantially between competitive and concentrated industries. In industries with HHI below the 50th percentile, technological complexity is associated with lower profitability, higher growth opportunities, and more intensive patent activity. Specifically, the coefficient of return on assets is  $-0.572$  and statistically significant, while the market-to-book ratio has a positive and highly significant coefficient of  $0.134$ . Patent growth is also positively associated with complexity, with a coefficient of  $0.055$ , while one-year citation counts remain negatively associated with complexity, with a coefficient of  $-0.075$ . These suggest that the relationship between complexity and financial structure is driven more by profitability and market valuation. Several mechanisms may explain these relationships in competitive industries. First, when companies have relatively similar market shares, technological differentiation can become an important way to gain competitive advantage. Greater patent activity and more complex combinations of technological knowledge may therefore emerge as firms attempt to distinguish themselves from competitors. Second, firms may strategically accumulate patents for defensive or offensive blocking purposes (Lerner (1995), and Blind, Cremers and Mueller (2009)), creating denser and more interconnected patent networks that increase measured technological complexity. Third, the negative association between ROA and technological complexity may reflect the short-run profitability costs of operating in technologically demanding industries. Developing and integrating complex technologies can require substantial experimentation, R&D investment, and organizational coordination, reducing contemporaneous profitability even when those investments strengthen firms' longer-run innovative capacity. Finally, the positive coeffi-

cient on the market-to-book ratio suggests that firms operating in industries with stronger growth opportunities and higher market valuations may have greater capacity to undertake risky and long-horizon technological investments. Investors may also incorporate the expected future cash flows and competitive advantages associated with technologically complex patent portfolios into market valuations.

By contrast, in industries with HHI at or above the 50th percentile, the pattern changes substantially. The size of the firm becomes positively associated with technological complexity: a 1% increase in total assets is associated with approximately a 0.013% increase in complexity. The market-to-book ratio becomes negative and significant, with a coefficient of  $-0.034$ , while the citation coefficient remains negative but substantially smaller in magnitude at  $-0.029$ . These results suggest that technological complexity in more concentrated industries may depend less on short-run innovation incentives and more on firms' scale and established technological capacity. Larger firms may have greater financial and technological resources to maintain broad and complex technological portfolios. At the same time, the negative market-to-book coefficient indicates that, among firms with greater market power, lower market valuations are associated with higher technological complexity. This pattern may reflect stronger incentives for relatively lower-valued firms in concentrated industries to pursue more complex innovation. The lower citation coefficient also suggests that technological influence is less closely connected to changes in network complexity in concentrated industries.

Robustness is further examined by splitting industries at the first and second tertiles of the Herfindahl–Hirschman Index to create a more extreme comparison using regression model 8. Table 6, columns (4) and (5), present these results. The contrast between competitive and concentrated industries becomes even stronger and the drivers remain the same as the median split results.

## 5 Related to Valuations

### 5.1 Data and Model

In this section, the industry technological complexity will be linked with market values to see how the market reaction is explained by the complexity score. In this section, various indicators of market value are used. The first used is the private economic value calculated in [Kogan et al. \(2017\)](#). The authors used the cumulative abnormal return around the patent announcement date to calculate the dollar value for all patents granted to public firms. By aggregating the KPSS value for all patents granted in each quarter, we get a firm-quarter-level indicator representing the dollar values assigned to the innovation activities during that quarter for the specific firm. The rest indicators are the quarterly BHAR using different types of abnormal return models.

The models used in this section is show in 9. The dependent variables are the market value indicators,  $V_{i,q}$  represents the value for firm  $i$  at quarter  $q$ . The main independent variable is the lagged industry technological complexity score  $C'_{k,q-1}$ , constructed from the CPC technology-class network and mapped to firms through their primary NAICS-4 industry  $k$ . Higher values of the score indicate that the industry's technological knowledge structure is more complex, diversified, and less concentrated within a simple technology cluster. To allow for a nonlinear relationship, the regressions also include the squared term of the lagged complexity score  $C_{k,q-1}^2$ .  $c(patent)_{i,q-1}^1$  is this firm's logged patent count at the previous quarter  $q - 1$ , and  $Citation_{i,q-13}^3$  is the logged number of citations the patent filed in quarter  $q - 13$  received within three years after filing, so that the citation counting stops at the end of quarter  $t - 1$  for patents filed at the end of quarter  $q - 13$ . Industry, year and quarter fixed effects are also added in the regression models. The linkage between patent and public firms follows the results from the updated dataset from [Kogan et al. \(2017\)](#).

$$V_{i,q} = \beta_{i,q}^0 + \beta_{i,q}^1 \cdot C_{k,q-1} + \beta_{i,t,q}^2 \cdot C_{k,q-1}^2 + \beta_{i,t}^p \cdot c(patent)_{i,q-1}^1 + \beta_{i,t}^C \cdot Citation_{i,q-13}^3 + \gamma_k + \alpha_t + \sigma_q + u_{i,t} \quad (9)$$

## 5.2 Results

Table 7 reports the relationship between industry-level technological complexity and firm-level market-value outcomes. The results show that the relationship between complexity and market outcomes is nonlinear. In Column (1), where the dependent variable is the KPSS patent-value measure, both the linear and squared terms of complexity are positive and statistically significant. This suggests a convex relationship between industry complexity and patent value: patents in more complex industries are associated with higher value, and this association becomes stronger at higher levels of complexity. One possible explanation is that highly complex technological environments generate more valuable innovation opportunities because successful inventions in these industries require broader knowledge integration and are harder to imitate.

By contrast, Columns (2)-(4), which use buy-and-hold abnormal returns based on the CAPM, FF3, and FF4 benchmarks, show a positive coefficient on the linear complexity term and a negative coefficient on the squared term. This pattern indicates an inverted-U-shaped relationship between complexity and stock-market reactions. At relatively low or moderate levels of complexity, higher industry complexity is associated with stronger abnormal returns, possibly because investors view complexity as a signal of richer technological opportunities and stronger innovation potential. However, when complexity becomes sufficiently high, the marginal effect turns negative. Meaning that, additional complexity may increase uncertainty, make the firm's technological trajectory harder to evaluate, lengthen commercialization timelines, and raise costs. Therefore, very high complexity may weaken the market reaction even if it is associated with valuable technological knowledge.

Overall, the results suggest that industry technological complexity is related to firm value, but the

direction of the relationship depends on the level of complexity and on the outcome being considered. Patent-value measures appear to reward high complexity more strongly, while stock-return outcomes display an inverted-U pattern, indicating that the market reacts positively to complexity only up to a certain point.

## **6 Conclusion**

Using patent data's CPC classification from 1985 to 2023, networks representing the knowledge component's connection can be built annually for each technology, and the structural diversity score measuring the heterogeneity degree of the network is proved to be better than other existing network metrics or complexity measurements in quantifying industrial complexity level.

This paper lays out a step-by-step method for converting complexity scores, patent counts, and citation counts from CPC to NAICS and between NAICS versions. The same procedure can be used to translate any other variables from CPC to NAICS or across different NAICS versions.

By digging into the industrial complexity score, this paper reveals that complexity rose steadily in all sectors until 2020. When the Covid-19 pandemic hit, complexity fell in labor-intensive industries – compared with the stable scores in tech-related areas, though employment didn't drop by the same amount. To make scores comparable across industries, we set each industry's own benchmark and calculated excess complexity. It has also been found that for some sectors, more affiliated industries do not have measurable complexity scores because they lack patents in the beginning. Our analysis also reveals clear contrasts in what drives complexity across industries. In competitive sectors, complexity depends mainly on patent activity – specifically, higher rates of new patents and fewer citations to older patents – however, in less competitive markets, only strong profits boost complexity.

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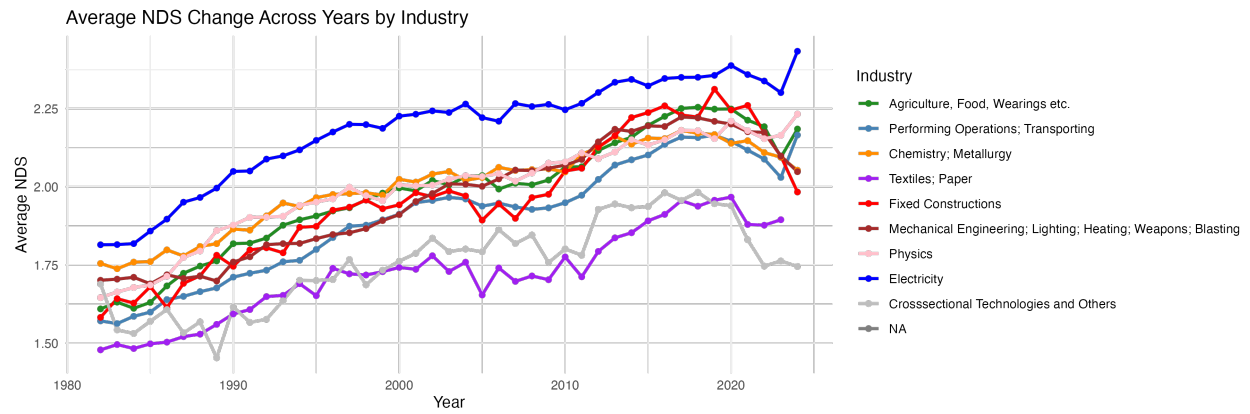


Figure 1: Complexity Score Across Years by 1-digit CPC Industry

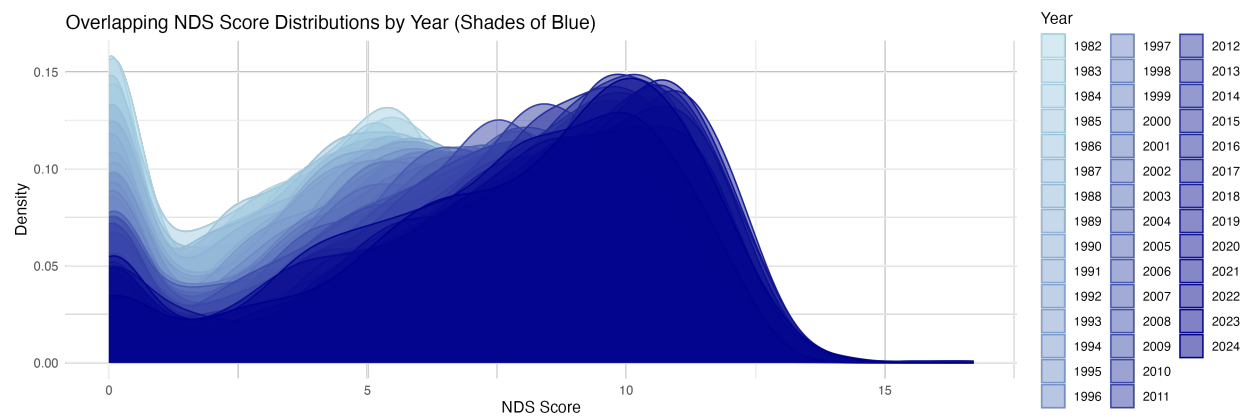


Figure 2: Overlapping NDS Score Distributions by Year

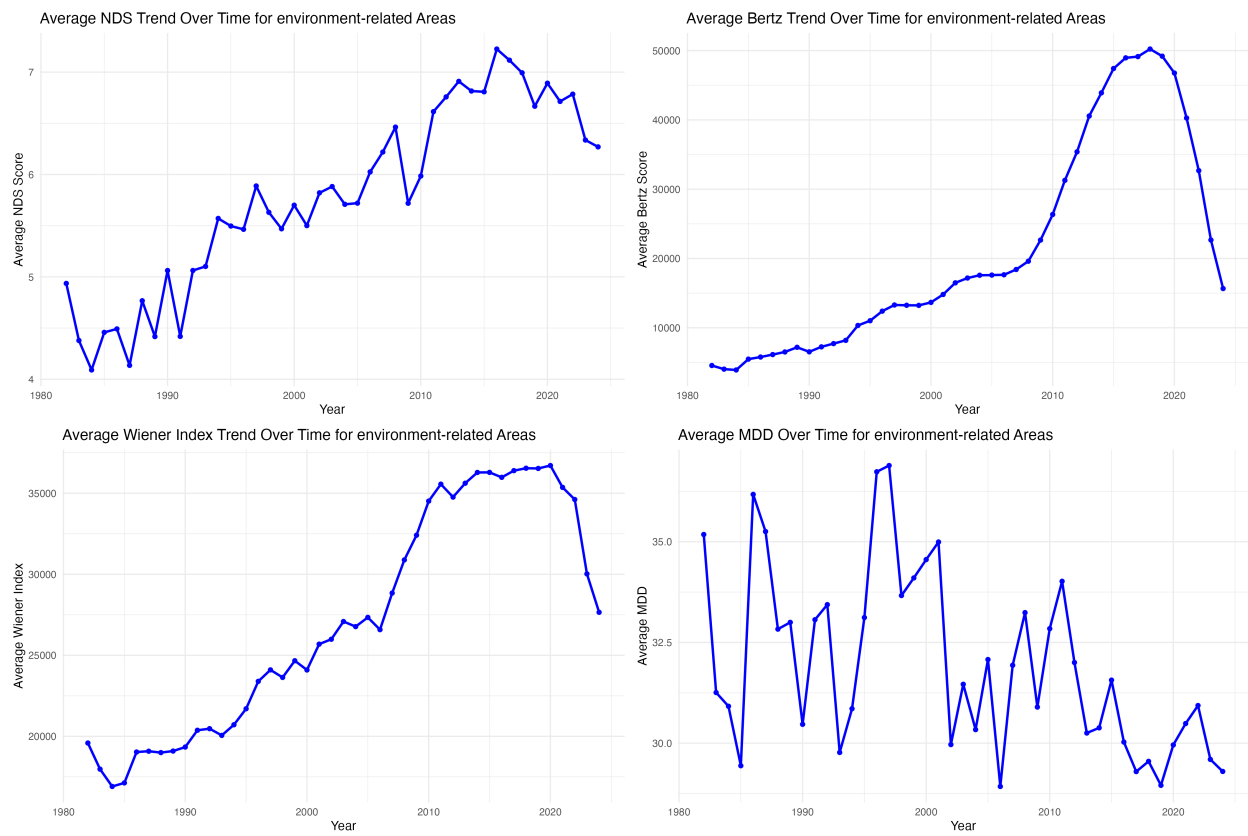


Figure 3: Annual Complexity Score of Environmental-related Areas

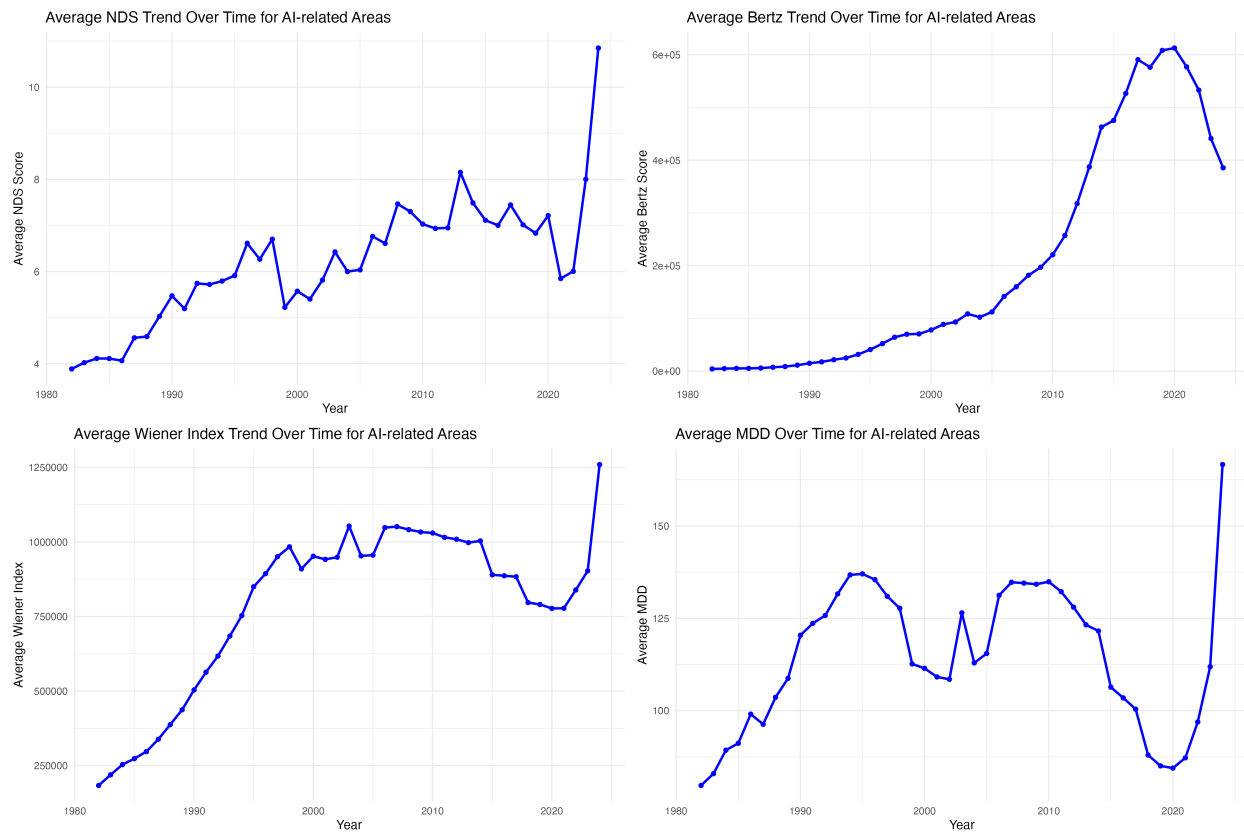


Figure 4: Annual Complexity Score of AI-related Areas

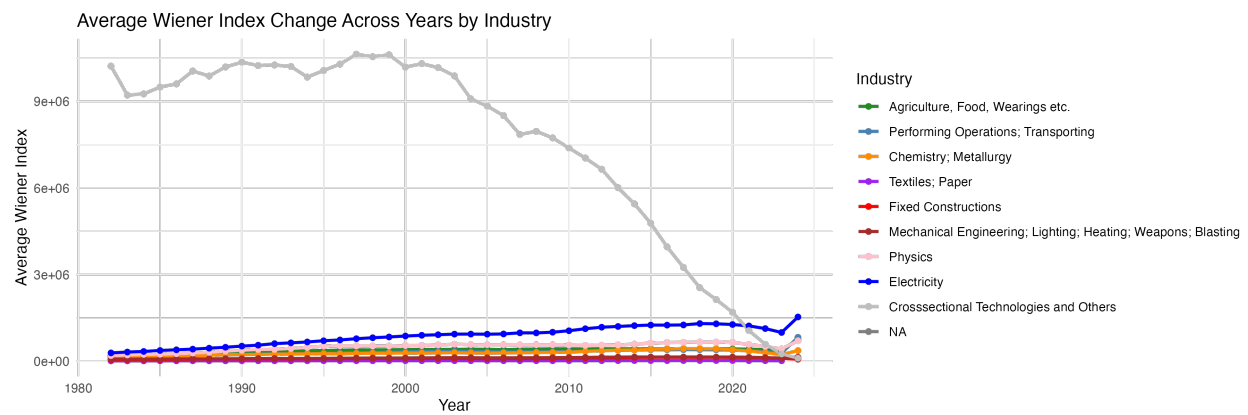


Figure 5: Complexity Wiener Index Across Years by 1-digit CPC Industry

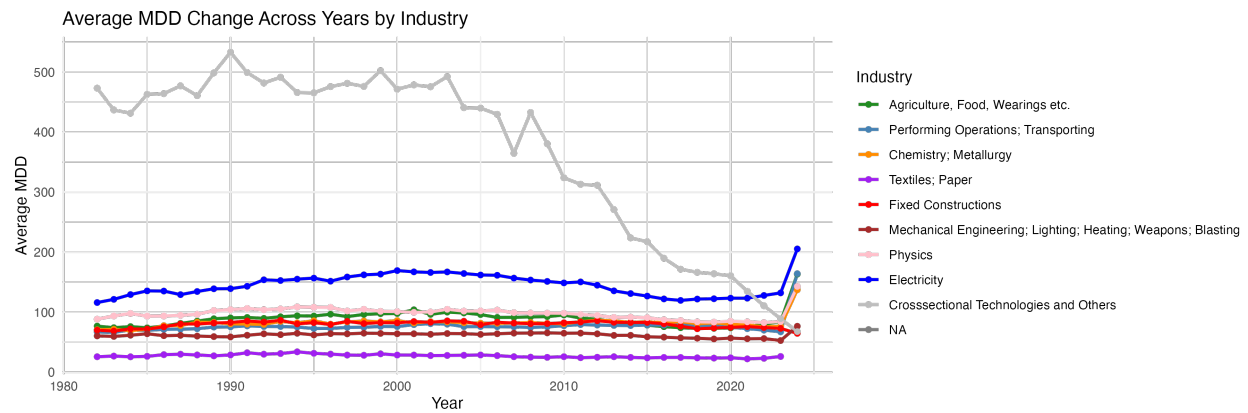


Figure 6: Complexity MDD Across Years by 1-digit CPC Industry

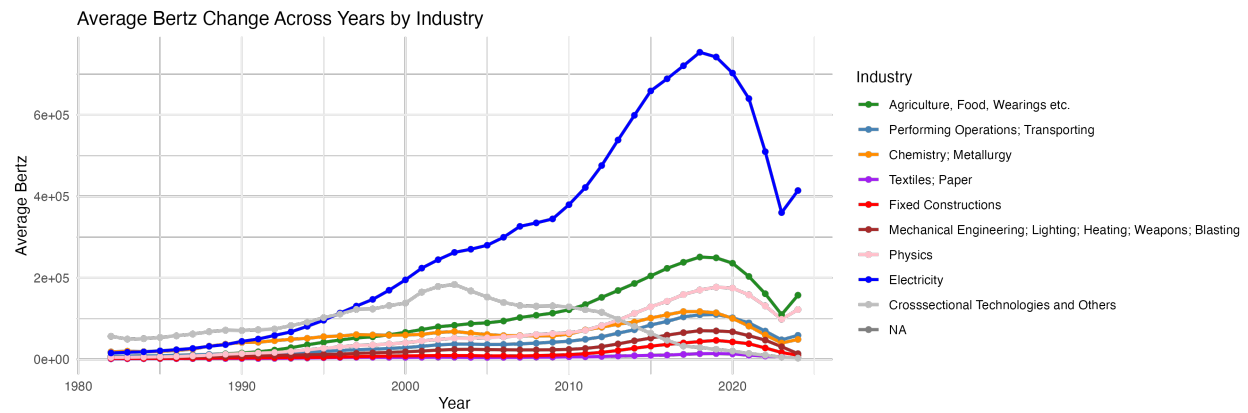


Figure 7: Complexity Bertz Index Across Years by 1-digit CPC Industry

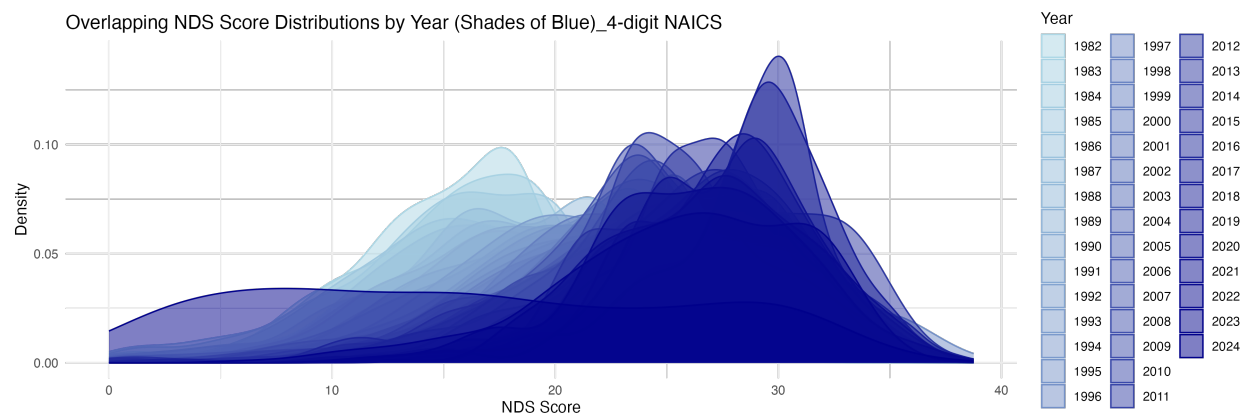


Figure 8: Overlapping NDS Score Distributions by Year at the level of 4-digit NAICS

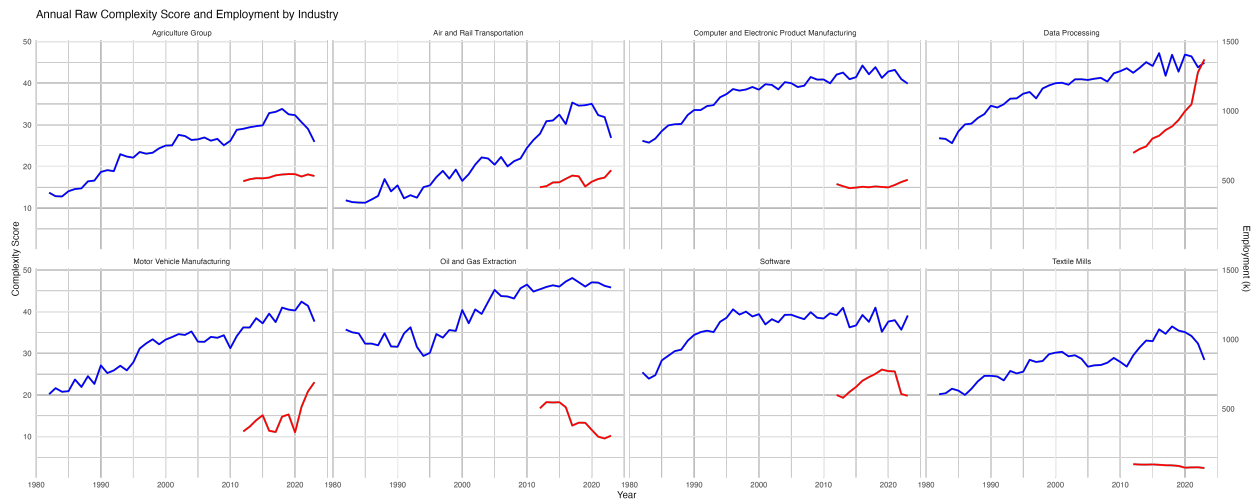


Figure 9: Annual Complexity Score and Employment by Industry

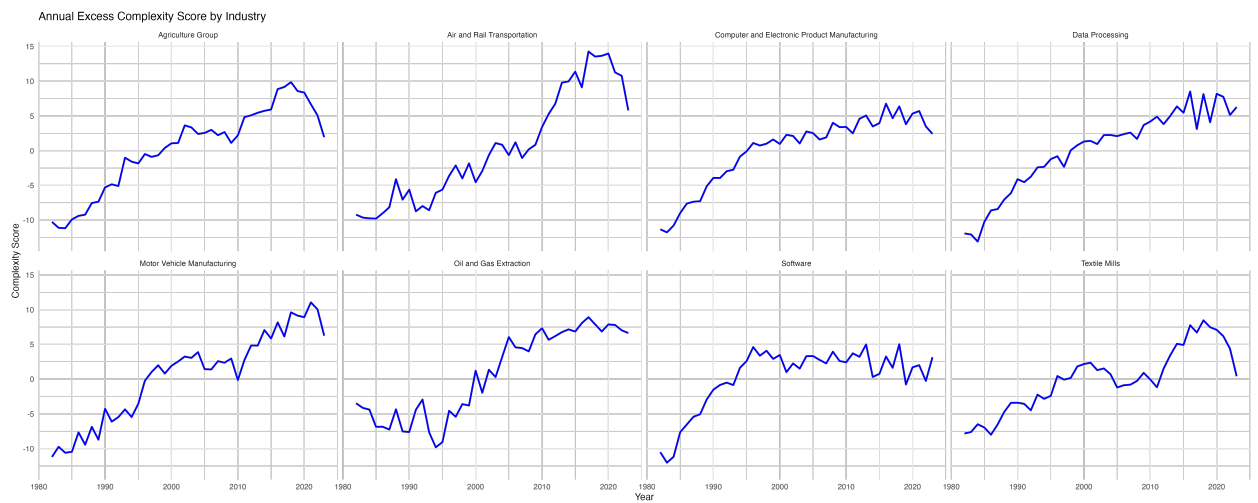


Figure 10: Annual Excess Complexity Score by Industry

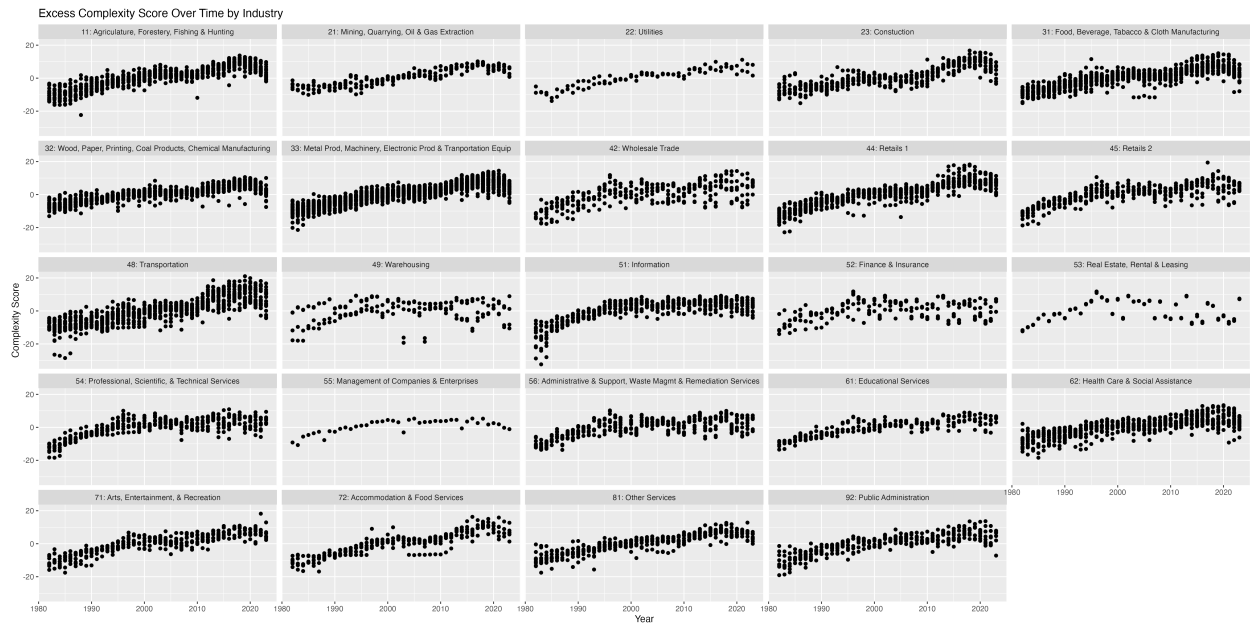


Figure 11: Annual Excess Complexity Score Over Time by Industry

Table 1: Statistics for Excess Score at 4-digit NAICS level

| 2-digit NAICS                                                      | Obs    | Count of 4-digit NAICS | Count w/ score data | Median | Std. dev. |
|--------------------------------------------------------------------|--------|------------------------|---------------------|--------|-----------|
| 11 (Agriculture, Forestry, Fishing & Hunting)                      | 773    | 19                     | 18                  | 0.83   | 6.57      |
| 21 (Mining, Quarrying, Oil & Gas Extraction)                       | 215    | 5                      | 5                   | 0.51   | 5.79      |
| 22 (Utilities)                                                     | 86     | 3                      | 2                   | 1.16   | 5.69      |
| 23 (Construction)                                                  | 430    | 10                     | 10                  | -0.47  | 6.57      |
| 31 (Food, Beverage, Tobacco & Cloth Manufacturing)                 | 938    | 22                     | 22                  | 0.59   | 5.95      |
| 32 (Wood, Paper, Printing, Coal Products, Chemical Manufacturing)  | 901    | 21                     | 21                  | 0.54   | 4.80      |
| 33 (Metal Prod, Machinery, Electronic Prod & Transportation Equip) | 1,803  | 43                     | 42                  | 0.58   | 6.11      |
| 42 (Wholesale Trade)                                               | 429    | 19                     | 10                  | 0.04   | 6.50      |
| 44 (Retail 1)                                                      | 686    | 16                     | 16                  | 0.69   | 7.07      |
| 45 (Retail 2)                                                      | 387    | 11                     | 9                   | 0.97   | 6.30      |
| 48 (Transportation)                                                | 902    | 25                     | 21                  | 0.07   | 7.66      |
| 49 (Postal Service & Warehousing)                                  | 171    | 4                      | 4                   | 1.57   | 6.61      |
| 51 (Information)                                                   | 473    | 12                     | 11                  | 1.75   | 6.76      |
| 52 (Finance & Insurance)                                           | 473    | 11                     | 11                  | -0.26  | 6.02      |
| 53 (Real Estate & Rental & Leasing)                                | 129    | 8                      | 3                   | -1.56  | 5.99      |
| 54 (Professional, Scientific & Technical Services)                 | 387    | 9                      | 9                   | 0.87   | 5.49      |
| 55 (Management of Companies & Enterprises)                         | 43     | 1                      | 1                   | 1.85   | 6.23      |
| 56 (Administrative & Support, Waste Mgmt & Remediation Service)    | 387    | 11                     | 9                   | 0.40   | 5.24      |
| 61 (Educational Services)                                          | 215    | 7                      | 5                   | 0.76   | 4.95      |
| 62 (Health Care & Social Assistance)                               | 687    | 18                     | 16                  | 0.83   | 5.78      |
| 71 (Arts, Entertainment, & Recreation)                             | 344    | 9                      | 8                   | -0.94  | 6.51      |
| 72 (Accommodation & Food Services)                                 | 258    | 7                      | 6                   | 1.14   | 7.27      |
| 81 (Other Services)                                                | 386    | 14                     | 9                   | 0.58   | 5.89      |
| 92 (Public Administration)                                         | 344    | 8                      | 8                   | 0.90   | 6.05      |
| 24 Sectors                                                         | 11,847 | 313                    | 276                 | 0.64   | 6.23      |

*Notes.* This table summarizes excess score statistics at the 2-digit NAICS level (the system's broadest category). Column 1 is the 2-digit NAICS sector. Column 2 reports, for each 2-digit NAICS sector, the total number of panel observations in the data set from 1985 to 2023. Column 3 shows the total count of 4-digit NAICS codes according to the Census Bureau, and Column 4 reports how many of those have valid excess scores over the same period. Finally, Columns 5 and 6 present the median and standard deviation of the excess scores for each sector.

Table 2: Definitions of Firm and Patent-Activity Variables

| Variable                   | Construction                             | Definition                                                                                                                                                                                       |
|----------------------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Log Total Assets           | $\ln(ATQ_{i,q-1})$                       | Natural logarithm of the firm's total assets, measured at the end of quarter $q - 1$ .                                                                                                           |
| Log Return on Assets       | $\ln(ROA_{i,q-1})$                       | Log-transformed return on assets, where $ROA_{i,q-1} = NIQ_{i,q-1}/ATQ_{i,q-1}$ . Net income and total assets are measured in quarter $q - 1$ .                                                  |
| Log Leverage Ratio         | $\ln(LVG_{i,q-1})$                       | Log-transformed leverage ratio, calculated as the sum of debt in current liabilities and long-term debt divided by current assets.                                                               |
| Log MB Ratio               | $\ln(MB_{i,q-1})$                        | Log-transformed MB ratio, calculated as market value divided by book value. Both are measured in quarter $q - 1$ .                                                                               |
| Herfindahl–Hirschman Index | $HHI_{j,q} = \sum_{f \in j} s_{f,j,q}^2$ | Industry concentration measured as the sum of squared market shares of firms operating in 4-digit NAICS industry $j$ during quarter $q$ . Higher values indicate greater industry concentration. |
| Patent Growth              | $g^1(\text{Patent})_{i,q-1}$             | Log-transformed growth rate of the patent count from quarter $q - 2$ to quarter $q - 1$ for industry $i$ .                                                                                       |
| One-year Citation Count    | $\text{Citation}_{i,q-5}^1$              | Log-transformed number of citations received within one year by patents filed by industry $i$ in quarter $q - 5$ . The one-year citation window ends in quarter $q - 1$ .                        |

*Notes.* The firm-level explanatory variables are lagged by one quarter. Patent growth is therefore measured through quarter  $q - 1$ . The citation variable is lagged by five quarters so that every patent has a complete one-year citation window before the outcome is measured in quarter  $q$ . For variables that can take negative values, the signed-log transformation is  $\text{sign}(x) \ln(1 + |x|)$ .

Table 3: Statistics for Accounting Variables and Patent Activity Indicators

| Name                                                         | Obs    | Mean        | P25       | Median     | P75        | Std. dev.   |
|--------------------------------------------------------------|--------|-------------|-----------|------------|------------|-------------|
| <i>Panel A: Reported in original units at industry level</i> |        |             |           |            |            |             |
| <i>Accounting Variables:</i>                                 |        |             |           |            |            |             |
| Total Assets (Mil.)                                          | 35,153 | 108,963.700 | 2,366.468 | 11,705.060 | 52,140.710 | 441,477.300 |
| ROA                                                          | 34,768 | -0.634      | -0.001    | 0.006      | 0.013      | 33.101      |
| Leverage Ratio                                               | 34,600 | 0.374       | 0.216     | 0.365      | 0.507      | 1.172       |
| MB ratio                                                     | 32,838 | 2.194       | 1.292     | 1.800      | 2.540      | 13.523      |
| HHI                                                          | 35,153 | 3,863.610   | 1,574.080 | 2,861.694  | 5,334.788  | 2,993.478   |
| <i>Patent Activity Indicators:</i>                           |        |             |           |            |            |             |
| Patent Growth                                                | 34,889 | 0.075       | -0.115    | 0.001      | 0.118      | 3.393       |
| 1-year Citation                                              | 35,153 | 177.445     | 0.200     | 4.086      | 65.116     | 643.042     |
| <i>Panel B: Reported in logged terms at industry level</i>   |        |             |           |            |            |             |
| <i>Accounting Variables:</i>                                 |        |             |           |            |            |             |
| Total Assets                                                 | 35,153 | 9.087       | 7.770     | 9.368      | 10.862     | 2.697       |
| ROA                                                          | 34,768 | -0.007      | -0.001    | 0.006      | 0.013      | 0.189       |
| Leverage Ratio                                               | 34,392 | 0.607       | 0.279     | 0.462      | 0.777      | 0.520       |
| MB ratio                                                     | 32,838 | 1.050       | 0.829     | 1.029      | 1.264      | 0.549       |
| HHI                                                          | 35,153 | 7.852       | 7.362     | 7.960      | 8.582      | 1.218       |
| <i>Patent Activity Indicators:</i>                           |        |             |           |            |            |             |
| Patent Growth                                                | 34,889 | -0.022      | -0.122    | 0.001      | 0.111      | 0.367       |
| 1-year Citation                                              | 35,153 | 2.363       | 0.182     | 1.626      | 4.191      | 2.344       |

*Notes.* This table summarizes the accounting variables and patent activity indicators at the industry level (4-digit NAICS). Panel A reports the summary statistics in their original units, while Panel B reports the corresponding natural-log-transformed variables. The log transformation is defined as  $\ln(x) = \text{sgn}(x) \ln(1 + |x|)$ .

Table 4: Robustness Checks for the Determinants of Industrial Technological Complexity

|                                                                             | (1)                  | (2)                  | (3)                  |
|-----------------------------------------------------------------------------|----------------------|----------------------|----------------------|
| <i>Panel A: Firm and industry characteristics</i>                           |                      |                      |                      |
| <i>Dependent variable: Industrial Technological Complexity Excess Score</i> |                      |                      |                      |
| Log Total Assets (lag 1)                                                    | -0.008**<br>(-2.03)  | -0.007<br>(-0.61)    | -0.008<br>(-0.66)    |
| Log Total Assets (lag 2)                                                    |                      | -0.004<br>(-0.38)    | -0.004<br>(-0.31)    |
| Log Total Assets (lag 3)                                                    |                      |                      | -0.003<br>(-0.23)    |
| Log Return on Assets (lag 1)                                                | -0.088***<br>(-3.34) | -0.088***<br>(-2.76) | -0.062*<br>(-1.84)   |
| Log Return on Assets (lag 2)                                                |                      | -0.020<br>(-0.61)    | -0.001<br>(-0.03)    |
| Log Return on Assets (lag 3)                                                |                      |                      | -0.038<br>(-1.14)    |
| Log Leverage Ratio (lag 1)                                                  | -0.014<br>(-1.09)    | -0.018<br>(-0.88)    | -0.014<br>(-0.63)    |
| Log Leverage Ratio (lag 2)                                                  |                      | -0.006<br>(-0.28)    | 0.009<br>(0.36)      |
| Log Leverage Ratio (lag 3)                                                  |                      |                      | -0.031<br>(-1.41)    |
| Log Market-to-Book Ratio (lag 1)                                            | -0.001<br>(-0.17)    | 0.025*<br>(1.93)     | 0.024*<br>(1.88)     |
| Log Market-to-Book Ratio (lag 2)                                            |                      | -0.029**<br>(-2.27)  | -0.028<br>(-1.64)    |
| Log Market-to-Book Ratio (lag 3)                                            |                      |                      | 0.002<br>(0.12)      |
| Log Herfindahl–Hirschman Index (lag 1)                                      | -0.014***<br>(-2.84) | -0.009<br>(-1.26)    | -0.009<br>(-1.19)    |
| Log Herfindahl–Hirschman Index (lag 2)                                      |                      | -0.007<br>(-1.01)    | -0.009<br>(-1.12)    |
| Log Herfindahl–Hirschman Index (lag 3)                                      |                      |                      | 0.001<br>(0.12)      |
| Industry fixed effects                                                      | Yes                  | Yes                  | Yes                  |
| Year fixed effects                                                          | Yes                  | Yes                  | Yes                  |
| Quarter fixed effects                                                       | Yes                  | Yes                  | Yes                  |
| <i>Panel B: Patent activity indicators</i>                                  |                      |                      |                      |
| <i>Dependent variable: Industrial Technological Complexity Excess Score</i> |                      |                      |                      |
| Log Patent Growth (lag 1)                                                   | 0.033*<br>(1.90)     | 0.059***<br>(3.00)   | 0.073***<br>(3.61)   |
| Log Patent Growth (lag 2)                                                   |                      | 0.054***<br>(2.74)   | 0.082***<br>(3.59)   |
| Log Patent Growth (lag 3)                                                   |                      |                      | 0.041*<br>(1.89)     |
| Log 1-year Citation (lag 5)                                                 | -0.056***<br>(-9.97) | -0.028***<br>(-4.14) | -0.019***<br>(-2.68) |
| Log 1-year Citation (lag 6)                                                 |                      | -0.054***<br>(-8.07) | -0.045***<br>(-6.36) |
| Log 1-year Citation (lag 7)                                                 |                      |                      | -0.026***<br>(-3.58) |
| Industry fixed effects                                                      | Yes                  | Yes                  | Yes                  |
| Year fixed effects                                                          | Yes                  | Yes                  | Yes                  |
| Quarter fixed effects                                                       | Yes                  | Yes                  | Yes                  |

*Notes.* This table reports separate robustness checks for the determinants of industrial technological complexity at the four-digit NAICS level. Panel A examines firm and industry characteristics, and Panel B examines patent activity indicators. Each set of three columns represents three separate regressions with the displayed lag structure. Column (1) includes the first displayed lag, column (2) adds the second displayed lag, and column (3) adds the third displayed lag. The variables are expressed in logarithmic terms. Parentheses report *t*-statistics. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.10$ .

Table 5: Correlations Between Accounting Variables and Patent-Activity Indicators

|               | Total Assets | ROA     | Lvg. Ratio | MB Ratio | HHI     | Patent Growth | 1-yr Citation |
|---------------|--------------|---------|------------|----------|---------|---------------|---------------|
| Total Assets  | 1.0000       |         |            |          |         |               |               |
| ROA           | 0.1141       | 1.0000  |            |          |         |               |               |
| Lvg. Ratio    | 0.0324       | -0.0637 | 1.0000     |          |         |               |               |
| MB Ratio      | 0.0292       | 0.0383  | -0.0662    | 1.0000   |         |               |               |
| HHI           | -0.6597      | -0.0800 | 0.0571     | 0.0442   | 1.0000  |               |               |
| Patent Growth | -0.0268      | 0.0031  | -0.0175    | -0.0007  | -0.0195 | 1.0000        |               |
| 1-yr Citation | 0.2712       | 0.0574  | -0.2012    | -0.0111  | -0.3017 | 0.0944        | 1.0000        |

*Notes.* This table reports the pairwise correlation matrix for the log-transformed accounting variables, industry characteristics, and patent-activity measures used in the analysis. The matrix is reported in lower-triangular form. Patent Growth is the log-transformed growth rate of the patent count from quarter  $q - 2$  to quarter  $q - 1$ . The 1-yr Citation variable is the log-transformed number of citations received within one year by patents filed in quarter  $q - 5$ ; its citation window ends in quarter  $q - 1$ .

Table 6: Regression Results for Determinants of Industry Technological Complexity

|                           | Log Excess Technological Complexity Score |                       |                       |                      |                       |
|---------------------------|-------------------------------------------|-----------------------|-----------------------|----------------------|-----------------------|
|                           | Full                                      | HHI Median Split      |                       | HHI Tertile Split    |                       |
|                           |                                           | < $p_{50}$            | $\geq p_{50}$         | < $p_{33}$           | > $p_{67}$            |
|                           | (1)                                       | (2)                   | (3)                   | (4)                  | (5)                   |
| Log Total Assets          | 0.008<br>(1.42)                           | 0.003<br>(0.23)       | 0.013**<br>(2.05)     | 0.021<br>(1.12)      | 0.021***<br>(2.62)    |
| Log Return on Assets      | -0.021<br>(-0.40)                         | -0.572*<br>(-1.91)    | 0.044<br>(0.80)       | -0.564**<br>(-2.01)  | 0.043<br>(0.82)       |
| Log Leverage Ratio        | -0.027<br>(-1.64)                         | -0.041<br>(-1.15)     | 0.001<br>(0.07)       | -0.032<br>(-0.73)    | 0.024<br>(1.19)       |
| Log Market-to-Book Ratio  | -0.004<br>(-0.45)                         | 0.134***<br>(5.07)    | -0.034***<br>(-3.79)  | 0.254***<br>(6.23)   | -0.050***<br>(-4.99)  |
| Log Patent Growth         | 0.038**<br>(2.03)                         | 0.055**<br>(1.99)     | 0.014<br>(0.62)       | 0.061*<br>(1.67)     | 0.017<br>(0.60)       |
| Log 1-year Citation Count | -0.064***<br>(-11.16)                     | -0.075***<br>(-8.96)  | -0.029***<br>(-3.63)  | -0.070***<br>(-6.38) | -0.013<br>(-1.21)     |
| Constant                  | -1.431***<br>(-27.67)                     | -1.462***<br>(-10.65) | -1.547***<br>(-26.32) | -1.789***<br>(-8.78) | -1.656***<br>(-24.79) |
| Industry FE               | Yes                                       | Yes                   | Yes                   | Yes                  | Yes                   |
| Year FE                   | Yes                                       | Yes                   | Yes                   | Yes                  | Yes                   |
| Quarter FE                | Yes                                       | Yes                   | Yes                   | Yes                  | Yes                   |
| Observations              | 31,077                                    | 16,245                | 14,832                | 10,476               | 9,002                 |
| Adjusted R-squared        | 0.556                                     | 0.540                 | 0.608                 | 0.540                | 0.620                 |

*Notes.* This table reports regression estimates based on Equation 8, with 4-digit NAICS industry fixed effects and year and quarter fixed effects. The dependent variable is the logarithm of the industry technological complexity score. Column (1) reports results for the full sample. Columns (2) and (3) divide industries according to whether their Herfindahl–Hirschman Index (HHI) is below or at or above the sample median. Columns (4) and (5) restrict the sample to industries in the bottom and top HHI tertiles, respectively; industries in the middle tertile are excluded. Firm-level explanatory variables and patent growth are lagged by one quarter. The one-year citation measure is lagged by five quarters so that each patent has a complete one-year citation window. Parentheses report heteroskedasticity-robust  $t$ -statistics. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.10$ .

Table 7: Technological Complexity and Market Valuations

|                           | Firms' Market Valuation |                     |                     |                     |
|---------------------------|-------------------------|---------------------|---------------------|---------------------|
|                           | KPSS<br>(1)             | BHAR CAPM<br>(2)    | BHAR FF3<br>(3)     | BHAR FF4<br>(4)     |
| Excess Complexity         | 0.033***<br>(3.09)      | 0.001***<br>(3.95)  | 0.003***<br>(11.12) | 0.003***<br>(11.16) |
| Excess Complexity Squared | 0.027***<br>(2.85)      | -0.001**<br>(-2.18) | -0.001**<br>(-2.02) | -0.000*<br>(-1.79)  |
| Patent Count              | 1.220***<br>(195.96)    | -0.001**<br>(-2.24) | -0.001<br>(-1.60)   | 0.000<br>(0.04)     |
| 3-year Firm Citations     | 0.122***<br>(33.75)     | 0.001***<br>(5.07)  | 0.001***<br>(5.08)  | 0.001***<br>(4.19)  |
| Industry FE               | Yes                     | Yes                 | Yes                 | Yes                 |
| Year FE                   | Yes                     | Yes                 | Yes                 | Yes                 |
| Quarter FE                | Yes                     | Yes                 | Yes                 | Yes                 |
| Observations              | 56,527                  | 657,035             | 657,035             | 651,530             |
| Adjusted R-squared        | 0.692                   | 0.022               | 0.010               | 0.008               |

*Notes:* This table reports regressions of firm outcomes on lagged industry-level excess technological complexity using the model 9. Column (1) uses  $\ln(\text{KPSS})$  as the dependent variable and includes only the firm-quarter observations with patent granted. Columns (2)-(4) use  $\ln(\text{BHAR})$  based on CAPM, FF3, and FF4 benchmarks, respectively, and all on-list firm-quarter observations are included. All specifications absorb NAICS-4, year and quarter fixed effects. Robust  $t$ -statistics are reported in parentheses. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.1$

# Data Appendix

## A Conversion from CPC to NAICS

### A.1 Conversion of Complexity Score

As mentioned in [Broekel \(2019\)](#), when calculating the NDS for the network of each 4-digit CPC code, 50 nodes are selected, and a subnetwork is drawn by a random walktrap of 150 steps starting from each of the node. These 50 subnetworks are used to calculate the iNDS and the final NDS is the average of the 50 iNDS. This process emphasizes the importance of topologies' diversity rather than the size of the network, so when the case is that multiple 4-digit CPC industries are mapped into one 6-digit NAICS industry, the scale complexity score should not be aggregated.

The probability weight needed for the conversion comes from [Nathan et al. \(2019\)](#) which utilized the 'Algorithmic Links with Probabilities' method and provided direct probabilistic linkages between classification systems and supports joint analysis of patent and economic data. The files that map 6-digit NAICS to 4-digit CPC were used, so for each 6-digit NAICS, the sum of probability weight across all the CPC codes it is mapped to is 1. The original score was calculated under each 6-digit NAICS code  $i$  might be mapped to  $n$  4-digit CPC code  $j$ s with the respective probability weight and original score for each  $j$  as  $Pr(j|i)$  and  $C_{CPC-4}^j$ . Then the 6-digit NAICS code's complexity score should be the sum of all the 4-digit CPC code's scores times the respective weight with the weight summing up to 1 as the equation 10.

$$C_{NAICS-6}^i = \sum_j C_{CPC-4}^j \cdot Pr(j|i) \quad \text{with} \quad \sum_j Pr(j|i) = 1 \quad (10)$$

### A.2 Conversion of Patent Count and Citation Data

After calculating the number of patents for each 4-digit CPC code in each year and the count of citations these patents receive in the subsequent one-, two-, and three- year periods, conversion to NAICS is needed for future analysis. The difference from the previous conversion is that the total

patent numbers and citation counts should be preserved after the crosswalk, so equation 10 cannot be applied to the patent and citation number.

As the complexity score, the original patent, or citation count is calculated under the 4-digit CPC level, a crosswalk to the 6-digit NAICS level is necessary for economic analysis. For this conversion, the files used mapped one or multiple NAICS industries to one CPC code, so the sum of weight assigned to these NAICS industries would be 1, and in consequence, the number of patents and citations does not change during the conversion.

For each 6-digit NAICS industry  $i$ , it may show up in the mapping into  $m$  4-digit CPC code  $j$ s with the respective probability weight for each  $i$  as  $Pr(j|i)$  and the patent or citation number as  $N_{CPC-4}^j$ . Then, the 6-digit NAICS code's patent or citation number should be the sum of all the 4-digit CPC code's patent or citation number, times the respective weight, as in equation 11.

$$N_{NAICS-6}^i = \sum_j N_{CPC-4}^j \cdot Pr(j|i) \quad (11)$$

## B Conversion between difference versions of NAICS

The crosswalks from CPC to NAICS are only feasible for the first three NAICS versions, which are the NAICS 1997, NAICS 2002, and NAICS 2007, so the conversion between different versions of NAICS is necessary between the recent 4 NAICS versions, which are NAICS 2007, NAICS 2012, NAICS 2017, and NAICS 2022. Once the conversion between different versions of NAICS are available, the conversion from CPC to the later versions of NAICS can be achieved.

### B.1 Conversion of Complexity Score

To convert complexity scores between NAICS versions, we use the Census Bureau's concordance files to assign probability weights. For instance, when mapping from 2007 to 2012 codes, all corresponding 6-digit NAICS 2007 entry is equally weighted across the 6-digit NAICS 2012 codes they map to. As a result, the weights for each 2012 code — summed over all its linked 2007 codes

— add up to 1. This ensures that when multiple 2007 codes merge into a single 2012 code, the aggregated complexity score remains on the same scale.

The complexity scores are first calculated or converted to one version. To move from one version to the next, we use equal probability weights from the Census concordance. Specifically, for each 6-digit code  $i$  in the new version that maps to  $m$  industries  $j$  in the old version, we give each link a weight of  $\frac{1}{m}$  with the respective complexity score for each  $j$  as  $C_{NAICS_n}^j$ . Then the new version's 6-digit NAICS complexity score is the average weighted of the old version's 6-digit NAICS complexity scores as the equation 12 .

$$C_{NAICS_{n+1}}^i = \frac{1}{m} \sum_j C_{NAICS_n}^j \quad (12)$$

## B.2 Conversion of Patent Count and Citation Data

As mentioned in Appendix A.2, we need to keep the total patent and citation counts unchanged during any version conversion, so we can't use Equation 12 for these variables.

The probability weights are also assigned by using the the Census Bureau's concordance files between two adjacent versions. To convert from old version to the new one, equal probability weights are assigned. Specifically, for each 6-digit code  $j$  in the old version, it may show up in the concordance with  $m$  industries  $i$  of the new version, we give each link a weight of  $Pr(j) = \frac{1}{m}$  with the respective patent count or citation count for each  $j$  as  $N_{NAICS_n}^j$ . The count for each new code  $i$  is then the sum of all linked old counts multiplied by their respective weights. This approach, shown in Equation 13, guarantees that the aggregate number of patents or citations stays the same through every step of the conversion process.

$$N_{NAICS_{n+1}}^i = \sum_j C_{NAICS_n}^j \cdot Pr(j) \quad (13)$$